CRYPTOGRAPHY AND NETWORK SECURITY LABORATORY	
Course Code: CYL57	**Credits: 0:0:1**
Prerequisite: Nil	**Contact Hours: 15P**
Course Coordinator:	

Course Contents

Part A

1. Develop a Java program to implement Caesar Cipher, Perform encryption and decryption with a user-defined shift value.
2. Develop a Java program to implement Playfair Cipher, encrypt a given plaintext using 5x5 matrix.
3. Develop a Java program to implement DES Algorithm. Simulate DES encryption and decryption using Java's security libraries or custom logic.
4. Write a Java program to implement AES Algorithm, use Java Cryptography Extension (JCE) to perform AES encryption and decryption with a 128-bit key.
5. Write a Java program to implement RSA Algorithm, generate public/private key pairs, encrypt a message with the public key, and decrypt using the private key.

Part B

1. Use Wireshark to capture and analyse SSL/TLS handshake. Perform a secure browser transaction and analyze packet-level handshake.
2. Capture and analyse HTTP/2 traffic over TLS using Wireshark. Open a modern website that supports HTTP/2 and inspect the protocol negotiation, multiplexed streams, and TLS encryption.
3. Initiate an SSH session between two local systems or virtual machines. Capture traffic using Wireshark and analyse the session key exchange, encryption setup, and user authentication.
4. Use a packet crafting tool such as Scapy (Python) to create custom TCP, UDP, or ICMP packets. Send these packets to a target system and observe how a firewall responds. Use Wireshark to capture and analyze the traffic, including flags, headers, and dropped packets.
5. Simulate an ARP spoofing attack in a controlled local network using tools like arpspoof or Ettercap. Capture the ARP traffic with Wireshark to observe the malicious ARP replies and analyze how the attacker intercepts traffic (Man-in-the-Middle).

Text Book:

1. "Cryptography and Network Security: Principles and Practice" by William Stallings, 7th Edition URL: https://williamstallings.com/Crypto/Crypto7e.html

Reference Book:

1. "Network Security Essentials: Applications and Standards" by William Stallings, 6th Edition, 2013

Course Outcomes (COs):
At the end of the course, the students should be able to:

1. Implement encryption and decryption algorithms to protect the network resources. (PO 1,2,3,4,5,10, PSO 1,2,3)
2. Apply key management and digital identity for digital signatures and demonstrate their utility in securing digital communication.(PO 1,2,3,4,5,10, PSO 1,2,3)
3. Demonstrate network security concepts to ensure secure transmission of information across communication networks(PO 1,2,3,4,5,10, PSO 1,2,3)

Course Assessment and Evaluation:

Parameter	**Marks**
CIE Test	20
Lab Record Writing + Viva+ program execution	30
Total	**50**
Final Exam will be conducted for 50 marks (SEE)	